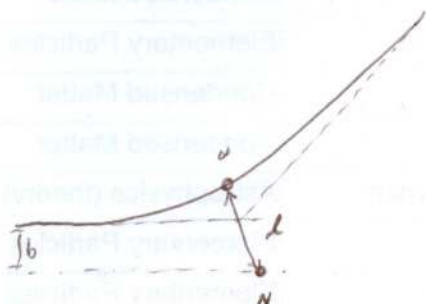


This formula is "almost" right, as we will see. It only agrees, as expected, if the α -particles & the nuclei ~~are~~ can still be regarded as classical particles.

~ Estimate of nucleus radius R_N



An upper bound on the size of a nucleus is given by the shortest distance an α -particle can approach to the nucleus l according to Rutherford's reasoning.

The shortest distance occurs for $b=0$ when

$$T = V$$

$$\Rightarrow \frac{1}{2} m v^2 = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{R_N l}$$

$$\Rightarrow l R_N = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{T} = \frac{Ze^2}{\pi\epsilon_0 m v^2}$$

Therefore, we conclude that $R_N \leq l(T_{\max})$, where $T_{\max}$ is the largest kinetic energy an incident particle can have.

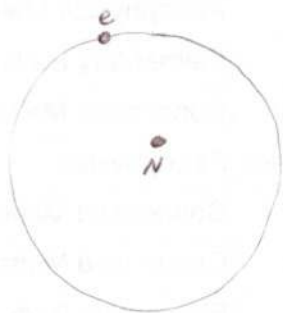
For a gold foil ($Z=79$) & the largest natural energy of an α -particle, $T_{\max} \sim 7.7 \text{ MeV} \sim 1.2 \times 10^{-12} \text{ J}$, we find

$$R_{Au} \leq l \approx 3 \times 10^{-14} \text{ m}$$

OBS: • $l \ll 10^{-10}$ typical atomic radius. $\rightarrow$ an atom is practically empty!
• The actual radius of the Gold nucleus is about $6 \times 10^{-15} \text{ m}$.

~ Atomic Orbits and Rutherford's failure

According to Rutherford, electrons describe orbits around the nucleus, much as a planetary system. A good estimate of the orbit radius is given by balancing the centrifugal & Coulomb forces assuming that the orbits are circular. Consider the Hydrogen:



$$\frac{1}{r} m v^2 = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

$$\Rightarrow v = \frac{e}{\sqrt{4\pi\epsilon_0 m r}}$$

The total energy E of an electron is then

$$E = T + V = \frac{1}{2}mv^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

$$= \frac{e^2}{8\pi\epsilon_0 r} - \frac{e^2}{4\pi\epsilon_0 r} = - \frac{e^2}{8\pi\epsilon_0 r} < 0 : \text{bound state}$$

OBS: If we added enough energy to the system p-e, such that $E \geq 0$, we would increase T so that the bound system would not be bound any longer. The exact amount to free an electron of an atom is called ionization energy. For ^1H , the ionization energy is 13.6 eV. $\sim 2.2 \times 10^{-18} \text{ J}$
 $\hookrightarrow$ can be measured

From $E = - \frac{e^2}{8\pi\epsilon_0 r}$ one can estimate the orbital radius

$$r = - \frac{e^2}{8\pi\epsilon_0 E} = \frac{(1.6 \times 10^{-19} \text{ C})^2 (9 \times 10^9 \text{ Nm}^2/\text{C}^2)}{2(-13.6 \text{ eV})}$$

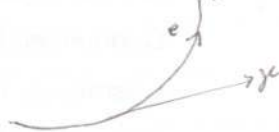
$\sim 2.2 \times 10^{-18} \text{ J}$

E.g. for ^1H

$$r \sim \frac{(1.6 \times 10^{-19} \text{ C})^2 (9 \times 10^9 \text{ Nm}^2/\text{C}^2)}{2(-2.2 \times 10^{-18} \text{ J})} \sim 5 \times 10^{-11} \text{ m}$$

Notice that $v \sim 2.2 \times 10^6 \text{ m/s}$ in this case.

Synchrotron radiation. According to electrodynamics, all accelerated charged particles emit a special kind of radiation: synchrotron radiation, losing thereby part of its energy.



Since As this happens, T will be reduced ~~NOT~~ destroying the ~~increasing~~ balance between the forces ~~observing~~. This will cause the atom to collapse in a spiral trajectory.

Although we see that atomic stability is compromised, one may be tempted to think that this problem would be solved by additional forces then unknown. There are however some aspects of the Rutherford assumptions that must be clarified. E.g. Rutherford assumed that α & H are classical particles. This is only valid if $\lambda \ll l$

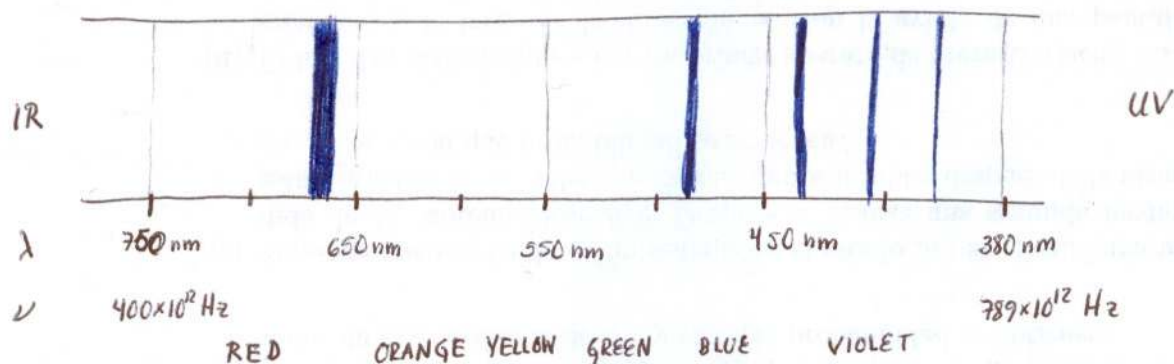
$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mT}} \sim 5 \times 10^{-15} \text{ m} \text{ for } \alpha \text{ particles with } T \sim 7.7 \text{ MeV}$$

$\rightarrow \lambda \ll l \text{ is satisfied!}$

~ Atomic spectra series

As we have seen, materials with $T \neq 0$ radiate in all frequencies of the em. spectrum, but exhibit different amplitudes at diff. freq.

In the XIX century it became known that even rarified gases emit radiation. Thus, the radiation should arise from isolated atoms. The distinctive characteristic of this radiation is that it's not a continuum. E.g. in the visible one observes that H emits in clearly defined energies



The same happens with any other element.

Curiously enough, elements exhibit also a discretized spectrum for the absorbed energies.

→ Elements absorb & emit in discrete intervals of ~~frequency~~

If Planck is right, then also only in discrete values of energy!

Balmer⁽¹⁸⁸⁵⁾ conceived a model in which the possible frequencies/ λ emitted by H were contained in the formula known as Balmer series:

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right) \quad n = 3, 4, 5, \dots$$

↳ Rydberg constant $R \sim 1.1 \times 10^7 \text{ m}^{-1} = 0.011 \text{ nm}^{-1}$

$$\Rightarrow \lambda \in \{ 656.3 \text{ nm}, 486.2 \text{ nm}, 434.1 \text{ nm}, 410.2 \text{ nm}, \dots, 364.6 \text{ nm} \}$$

$n = 3, \quad 4, \quad 5, \quad 6, \quad \dots \quad \infty$

Similar series were found in the UV

Lyman $\frac{1}{\lambda} = R \left(\frac{1}{1^2} - \frac{1}{n^2} \right) \quad n=2, 3, 4, \dots \quad \lambda \in \left\{ \underset{2}{121.5}, \underset{3}{102.5}, \underset{4}{97.2}, \dots, \underset{\infty}{91.2} \right\}$

& in the IR

Paschen $\frac{1}{\lambda} = R \left(\frac{1}{3^2} - \frac{1}{n^2} \right) \quad n=4, 5, 6, \dots \quad \lambda \in \left\{ \underset{4}{1875.2}, \underset{5}{1281.9}, \underset{6}{1093.9}, \dots, \underset{\infty}{820.4} \right\}$

Brackett $\frac{1}{\lambda} = R \left(\frac{1}{4^2} - \frac{1}{n^2} \right) \quad n=5, 6, 7, \dots \quad \lambda \in \left\{ \underset{5}{4051.4}, \underset{6}{2625.3}, \underset{7}{2165.7}, \dots, \underset{\infty}{1458.5} \right\}$

Pfund $\frac{1}{\lambda} = R \left(\frac{1}{5^2} - \frac{1}{n^2} \right) \quad n=6, 7, 8, \dots \quad \lambda \in \left\{ \underset{6}{7458.4}, \underset{7}{4652.8}, \underset{8}{3739.8}, \dots, \underset{\infty}{2278.9} \right\}$

Similar series were also found for other elements.

Rutherford's model cannot describe these results. Any successful model should reproduce these series.

~ Bohr's model

From the balance between the centripetal & Coulomb forces, we found that for H the electron speed is given by

$$v = \frac{e}{\sqrt{4\pi\epsilon_0 m r}}$$

$$\Rightarrow \lambda = \frac{h}{mv} = \frac{h}{e} \sqrt{\frac{4\pi\epsilon_0 r}{m}}$$

The radius of the (smallest) H orbit was found to be $r = 5.3 \times 10^{-11} \text{ m}$

$$\rightarrow \lambda \approx 33 \times 10^{-11} \text{ m}$$

which curiously coincides with the circumference of the orbit

$$2\pi r \approx 33 \times 10^{-11} \text{ m} = \lambda !$$

In Bohr's model, this is a clear hint that, independently of the kind of atom, the circumference of an orbit must coincide with a multiple of λ .

Bohr thought that this was most natural because, otherwise the wave associated to the electron would exhibit destructive interferences

Bohr's conjecture is expressed as

$$2\pi r_n = n\lambda \quad (\Rightarrow L = n\hbar)$$

where we notice that the smallest possible electron orbit is circular while others may adopt larger different shapes



We also notice that it's possible to obtain a precise value for r_n provided our expression of λ

$$2\pi r_n = n \frac{h}{e} \sqrt{\frac{4\pi\epsilon_0}{m}} \quad \Leftrightarrow \quad r_n = n^2 \frac{h^2 \epsilon_0}{\pi m e^2} \equiv n^2 a_0 \quad \text{Bohr radius}$$

→ radius quantization

$$a_0 = 5.3 \times 10^{-11} \text{ m}$$

Further, since the electron energies depend on r as

$$E = - \frac{e^2}{8\pi\epsilon_0 r}$$

we also find a quantization of the energy levels

$$E_n = - \frac{1}{n^2} \frac{me^4}{8\epsilon_0^2 \hbar^2} \equiv + \frac{1}{n^2} E_1$$

$$E_1 = -13.6 \text{ eV} \quad \text{energy of ground state}$$

$$E_n \quad n > 1 \quad \text{energy of excited states}$$

In Bohr's model E_1 corresponds to the ionization energy of H.

Similar expressions are obtained for other elements and can thus be contrasted with the actual ionization energies found experimentally (see figure)

a direct comparison of emission spectral series leads to the following observations

$$R = - \frac{E_1}{hc} \rightarrow \text{Rydberg's constant is not fundamental!}$$

Lyman	$n_f = 1$	...	UV
Balmer	$n_f = 2$	...	Visible
Paschen	$n_f = 3$	}	IR
Bracket	$n_f = 4$		
Pfund	$n_f = 5$		

Clearly, it's trivial to extend these results to any element with atomic number Z

$$\frac{1}{\lambda} = - \underbrace{\frac{Z^2 E_1}{hc}}_{\equiv R_Z} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$n_f < n_i$: emission
 $n_i < n_f$: absorption
 E_1 : ground-state energy of H
 R_Z : Rydberg's constant for atom with Z

~ Corrections to Bohr's model

Bohr's model was contested by many scientists (e.g. theoreticians Otto Stern & Max von Laue said they'd quit physics if Bohr was right; they didn't!). However, the crucial problems were:

1. Why can't all electrons occupy the ground state? (see Pauli)
2. Why were there tiny discrepancies in R and λ wrt measured values?

The first question was answered by Pauli with his exclusion principle, which shall be studied later. The latter had a simpler answer in terms of the reduced mass of the electron.

Since the nucleus has mass M and it's moving, the motion of the electron is affected. In fact, the motion of the smallest particle in such a system is given ~~by~~ in terms of

$$m' = \frac{mM}{m+M} < m \quad \text{reduced mass}$$

Therefore the corrected energy levels should be

$$E_n' = -\frac{m'e^4}{8\epsilon_0^2 h^2} \frac{1}{n^2} = -\frac{m'}{m} \frac{me^4}{8\epsilon_0^2 h^2} \frac{1}{n^2} = \frac{m'}{m} \frac{E_1}{n^2}$$

Since $\frac{m'}{m} \approx 0.9995$ for H

then the energy levels are down-shifted by a 0.05%
where-as the λ 's are up-shifted by the same factor ($\lambda \propto \frac{1}{E}$)

Possible extensions (not compulsory)

Hydrogen-like atoms

Mu: $\mu^+ - e^-$

Ps: $e^+ - e^-$

deuterium

~ Positronium, muonium, deuterium

As a generalization of the hydrogen description, it is possible to study similar atoms. As an example consider the positronium^(Ps), consisting of a positron around which an electron orbits. In this case

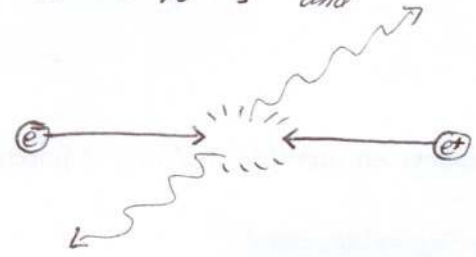
$$m_e' = \frac{m_e m_{e^+}}{m_e + m_{e^+}} = \frac{m_e^2}{2m_e} = \frac{m_e}{2}$$

This, we expect a correction to the energy spectrum by a large factor

$$\frac{m_e'}{m_e} = \frac{1}{2}$$

$$\Rightarrow E_n' = \frac{1}{2} \frac{E_1}{n^2} < E_n, \quad \frac{1}{\lambda'} = -\frac{1}{2} \frac{E_1}{hc} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right), \quad r_n' = \frac{m_e}{m_e'} n^2 a_0 = 2n^2 a_0$$

The Ps was predicted in 1932-1934 and discovered @ MIT in 1951. Its lifetime is about $10^{-10} - 10^{-7}$ s and



* Another exotic atom is the muonium Mu, consisting of a μ^+ & e^-

$$m_e' = \frac{m_\mu m_e}{m_\mu + m_e} = \frac{207 m_e}{208} \approx 0.9952 m_e$$

$$\Rightarrow E_n' \approx 99.52\% E_n, \quad \lambda' \approx 1.0048 \lambda, \quad r_n' \approx 1.0048 r_n$$

* Also the deuterium can be studied in this context. Its nucleus consists of a proton & a neutron around which an electron orbits.

$$m_e' = \frac{m_e (1836 + 1838) m_e}{(1836 + 1838) m_e + m_e} \approx 99.97\% m_e$$

$$\Rightarrow E_n' \approx 99.97\% E_n, \quad \lambda' \approx 1.0003 \lambda, \quad r_n' \approx 1.0003 r_n$$

~ Correspondence principle

The correspondence principle was introduced by Bohr several years after he first used when establishing the properties of his atomic model. It states a connection between the quantum universe & the classical description.

One form to enunciate this principle is:

"The behavior of systems described by quantum mechanics reproduces classical physics in the limit of large quantum numbers."

We know already an example of this correspondence; the classical atomic model of Rutherford's is a correct description because in this case

$$\lambda \ll d \quad \text{de Broglie } \alpha\text{-wavelength} \quad \text{minimal distance between } \alpha \text{ \& } N$$

An example of application of the correspondence principle is the synchrotron radiation emitted by a large atom. Consider e.g. an atom with $r_n = 0.01 \text{ mm}$, then

$$r_n = a_0 n^2 \Leftrightarrow n = \sqrt{\frac{r_n}{a_0}} \approx \sqrt{\frac{10^{-5} \text{ m}}{5.3 \times 10^{-11} \text{ m}}} \sim 435 \quad \text{Large!}$$

For such an ^{large} atom, with large quantum number n , its observable properties should also be described by classical mechanics.

Let's study the synchrotron radiation. The frequency at which such radiation is emitted is given by the frequency of revolution & multiples thereof:

$$f = \frac{v}{2\pi r}$$

From our previous treatment we know that

$$v = \frac{e}{\sqrt{4\pi\epsilon_0 m_e r}} \quad \& \quad r_n = a_0 n^2 = \frac{h^2 \epsilon_0}{\pi m_e e^2} n^2$$

$$\Rightarrow f = \frac{e}{2\pi \sqrt{4\pi\epsilon_0 m_e r^3}} = \frac{e}{2\pi \sqrt{4\pi\epsilon_0 m_e} \sqrt{\frac{h^2 \epsilon_0}{\pi m_e e^2} n^2}} \frac{e^3}{h^3 n^3} = \frac{m_e^4}{8\epsilon_0^2 h^3} \frac{2}{n^3} = -\frac{E_1}{h} \frac{2}{n^3}$$

If the correspondence principle holds, this frequency must coincide with the emitted-photon frequency ν for large n , in transitions to close lower E -levels:

$$\nu = -\frac{E_1}{h} \left(\frac{1}{n^2} - \frac{1}{n^2} \right) = -\frac{E_1}{h} \left(\frac{1}{(n-p)^2} - \frac{1}{n^2} \right) \quad p \ll n$$

$$= -\frac{E_1}{h} \left(\frac{2pn + p^2}{n^2(n-p)^2} \right)$$

$$\approx -\frac{E_1}{h} \frac{2pn}{n^4} = -\frac{E_1}{h} \frac{2p}{n^3} \quad \text{for large } n$$

We see that for $p=1$, this equation exactly coincides with f . If p increases, we obtain harmonics of f , as also happens for synchrotron rad.

For small n , the exact expression of ν , the quotient $\frac{\nu}{f} \sim \mathcal{O}(2-3)$, but for large n , classical & quantum are equal.